```
clc;
clear;
close all;
```

1. SYSTEM PARAMETERS

```
totalBandwidthHz = 1e6;
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7;
```

2. CAPACITY CALCULATIONS

```
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);

tdmaUsers = tdmaSlots;

targetSIR = 10^(targetSIRdB / 10);

cdmaUsers = floor(1 + processingGain / targetSIR);
```

3. DISPLAY RESULTS

```
fprintf('\n=======================================\n');
fprintf(' FDMA, TDMA AND CDMA COMPARISON\n');
fprintf('=======================================\n');

fprintf('Total Bandwidth = %.2f MHz\n',totalBandwidthHz/1e6);
fprintf('FDMA Channel Bandwidth = %.2f kHz\n',fdmaChannelHz/1e3);
fprintf('TDMA Slots per Frame = %d\n',tdmaSlots);
fprintf('CDMA Processing Gain = %d\n',processingGain);
fprintf('Target SIR = %.2f dB\n',targetSIRdB);

fprintf('\nILLUSTRATIVE USER CAPACITY\n');
fprintf('FDMA = %d users\n',fdmaUsers);
fprintf('TDMA = %d users\n',tdmaUsers);
fprintf('CDMA = %d users\n',cdmaUsers);
```

```
=======================================
 FDMA, TDMA AND CDMA COMPARISON
=======================================
Total Bandwidth = 1.00 MHz
FDMA Channel Bandwidth = 200.00 kHz
TDMA Slots per Frame = 8
CDMA Processing Gain = 64
Target SIR = 7.00 dB

ILLUSTRATIVE USER CAPACITY
```

```
FDMA = 5 users
TDMA = 8 users
CDMA = 13 users
```

4. COMPARISON TABLE

```
Method = {'FDMA';'TDMA';'CDMA'};

SupportedUsers = [
    fdmaUsers
    tdmaUsers
    cdmaUsers
    ];

results = table(Method,SupportedUsers);

fprintf('\nCAPACITY COMPARISON TABLE\n');
disp(results);
```

```
CAPACITY COMPARISON TABLE
     Method     SupportedUsers
    ________    ______________

    {'FDMA'}           5
    {'TDMA'}           8
    {'CDMA'}          13
```

5. CDMA SIR CALCULATION

```
K = 2:30;

sirLinear = processingGain ./ (K-1);
sirDb = 10*log10(sirLinear);
```

6. GRAPH 1: CAPACITY COMPARISON

```
figure;

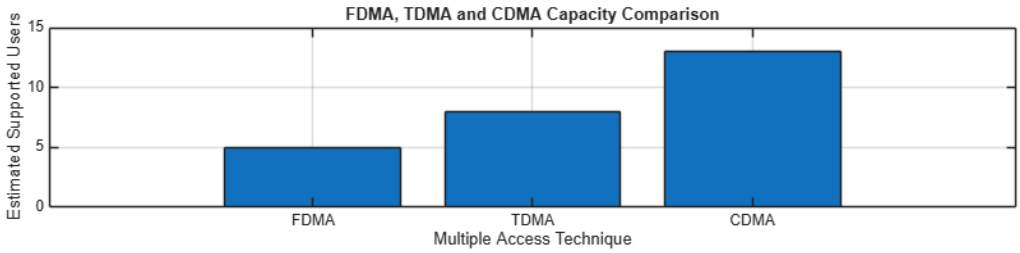
bar(SupportedUsers);

set(gca,'XTick',1:3);
set(gca,'XTickLabel',Method);

xlabel('Multiple Access Technique');
ylabel('Estimated Supported Users');

title('FDMA, TDMA and CDMA Capacity Comparison');

grid on;
```



7. GRAPH 2: CDMA SIR VS NUMBER OF USERS

```
figure;

plot(K,sirDb,'bo-','LineWidth',1.5,'MarkerSize',5);
hold on;

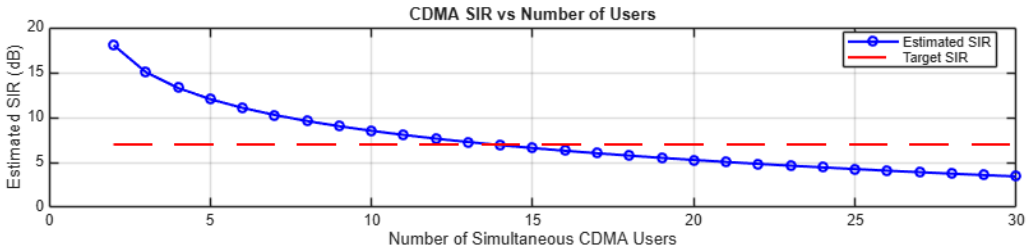
plot([min(K) max(K)], ...
    [targetSIRdB targetSIRdB],'r--','LineWidth',1.5);

xlabel('Number of Simultaneous CDMA Users');
ylabel('Estimated SIR (dB)');

title('CDMA SIR vs Number of Users');

legend('Estimated SIR','Target SIR','Location','best');

grid on;
hold off;
```



8. FINAL RESULT

```
fprintf('\n=======================================\n');
fprintf(' FINAL RESULT\n');
fprintf('=======================================\n');

fprintf('1. FDMA capacity estimate = %d users.\n',fdmaUsers);
fprintf('2. TDMA capacity estimate = %d users.\n',tdmaUsers);
fprintf('3. CDMA capacity estimate = %d users.\n',cdmaUsers);
fprintf('4. CDMA SIR decreases as simultaneous users increase.\n');
fprintf('5. Both comparison graphs generated successfully.\n');
```

```
=======================================
 FINAL RESULT
=======================================
1. FDMA capacity estimate = 5 users.
2. TDMA capacity estimate = 8 users.
3. CDMA capacity estimate = 13 users.
4. CDMA SIR decreases as simultaneous users increase.
5. Both comparison graphs generated successfully.
```

Published with MATLAB® R2026b